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5.6.3 Sizing for Critical Flow

5.6.3.1 General

5.6.3.1.1 Pressure-relief devices in gas or vapor service that operate at critical flow conditions (see 5.6.2) may be sized using Equation (2) through Equation (7). Each of the equations may be used to calculate the effective discharge area, AA , required

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to achieve a required flow rate through a pressure-relief device. A PRV that has an effective discharge area equal to or greater than the calculated value of AA is then chosen for the application.

In USC units:

$$A = \frac{W}{C K_d P_1 K_b K_c} \sqrt{\frac{T Z}{M}}$$

$$A = \frac{W}{C K_d P_1 K_b K_c} \sqrt{\frac{T Z}{M}} \quad \text{tag {2}}$$

$$A = \frac{V \sqrt{T Z M}}{6.32 C K_d P_1 K_b K_c}$$

$$A = \frac{V \sqrt{T Z M}}{6.32 C K_d P_1 K_b K_c} \quad \text{tag {3}}$$

$$A = \frac{V \sqrt{T Z G_v}}{1.175 C K_d P_1 K_b K_c}$$

$$A = \frac{V \sqrt{T Z G_v}}{1.175 C K_d P_1 K_b K_c} \quad \text{tag {4}}$$

In SI units:

$$A = \frac{W}{C K_d P_1 K_b K_c} \sqrt{\frac{T Z}{M}}$$

$$A = \frac{W}{C K_d P_1 K_b K_c} \sqrt{\frac{T Z}{M}} \quad \text{tag {5}}$$

$$A = \frac{2.676 \times V \sqrt{T Z M}}{C K_d P_1 K_b K_c}$$

$$A = \frac{2.676 \times V \sqrt{T Z M}}{C K_d P_1 K_b K_c} \quad \text{tag {6}}$$

$$A = \frac{14.41 \times V \sqrt{T Z G_v}}{C K_d K_1 P K_b K_c}$$

$$A = \frac{14.41 \times V \sqrt{T Z G_v}}{C K_d K_1 P K_b K_c} \tag{7}$$

where

A

A is the required effective discharge area of the device, in. ² (mm²)

(mm²) (see 3.20);

W is the required flow through the device, lb/h (kg/h);

C is a function of the ratio of the ideal gas specific heats ($k = C_p / C_v$) ($k = C_{\text{p}} / C_{\text{v}}$) of the gas or vapor at inlet relieving temperature.

The coefficient, C , is determined as follows.

In USC units [for use in Equation (2) through Equation (4) only]:

$$C = 520 \sqrt{k \left(\frac{2}{k+1} \right)^{\frac{(k+1)}{(k-1)}}}$$

$$C = 520 \sqrt{k \left(\frac{2}{k+1} \right)^{\frac{(k+1)}{(k-1)}}} \tag{8}$$

In SI units [for use in Equation (5) through Equation (7) only]: